
Geometric Representation Theory

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

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first chap

1.1 definitions

Definition 1.1.1: Algebraic Group

Let G be a variety. Then G is an *algebraic group* over an algebraically closed field of characteristic 0 if it has a group structure with multiplication and inversion:

$$m : G \times G \rightarrow G \quad \text{and} \quad \iota : G \rightarrow G$$

that are also morphisms of varieties. No harm in assuming $k = \mathbb{C}$ by the Lefschetz principle.

Note that G is *not* a topological group (except in dimension 0) as $G \times G$ has the *Zariski topology* and not the product topology (show a topological group that is T_1 must be T_2 , but G is not T_2). The translation map $x \mapsto xy$ is always an isomorphism, which also means that all geometric properties points are identical. As there are always smooth points, all points of G must be smooth.

Chevalley splits algebraic groups into “affine” and “abelian”. Namely G has a normal subgroup H such that H is affine and G/H is projective (i.e. an “abelian variety”).

We’ll focus on affine algebraic groups so $G = \text{Spec } A$. So that $G \times G \rightarrow G$ corresponds to $\Delta : A \rightarrow A \otimes A$ that makes A into a *Hopf algebra*. and:

1. Δ is co-associative
2. there exists a co-unit $\epsilon : A \rightarrow k$ satisfying the usual axiom of a ring, and
3. there exists an inversion $\iota : A \rightarrow A$.

Example 1.1.2: Algebraic Groups

1. $G = \mathbb{G}_a = \mathbb{A}^1$.

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